NERVOUS CONTROL OF GIT

Dr Shahneela Siraj
Associate Professor
DIMC,DUHS.

- The gastrointestinal tract is regulated, in part, by the autonomic nervous system, which has an extrinsic component and an intrinsic component.
- The extrinsic component is the sympathetic and parasympathetic innervation of the gastrointestinal tract.
- The intrinsic component is called the enteric nervous system.

INTRINSIC NERVE SUPPLY – ENTERIC NERVOUS SYSTEM

- It lies entirely in the wall of the gut, beginning in the esophagus and extending all the way to the anus. The number of neurons in this enteric system is about 100 million, nearly equal to the number in the entire spinal cord. This highly developed enteric nervous system is especially important in controlling gastrointestinal movements and secretion.

- The enteric nervous system is composed mainly of two plexuses,
- (1) an outer plexus lying between the longitudinal and circular muscle layers, called the **myenteric plexus or Auerbach's plexus**, &
- (2) an inner plexus, called the **submucosal plexus or Meissner's plexus**, which lies in the submucosa.
- The intrinsic or enteric nervous system can direct all functions of the gastrointestinal tract, even in the absence of extrinsic innervation.

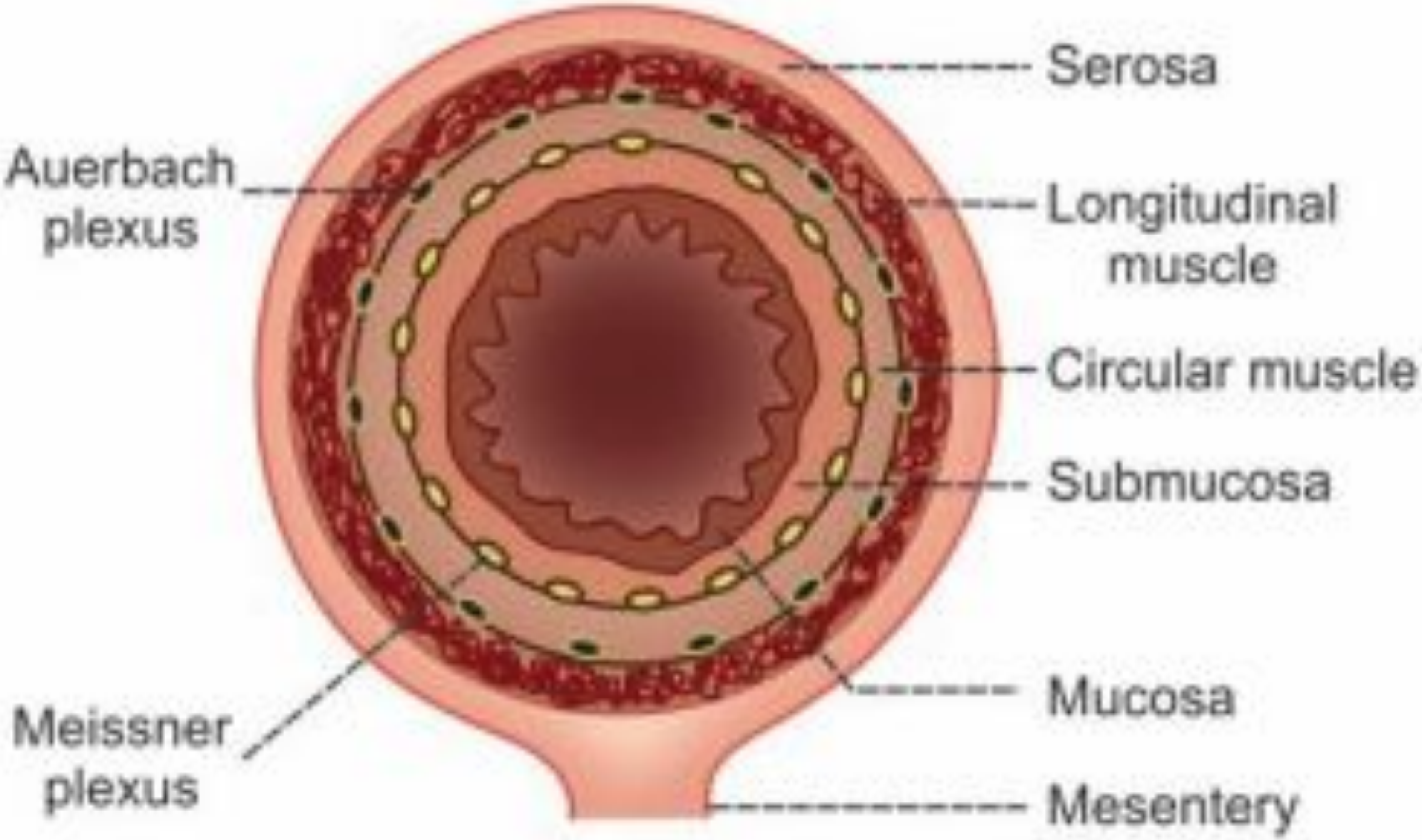

FIGURE 36.2: Structure of intestinal wall with intrinsic nerve plexus

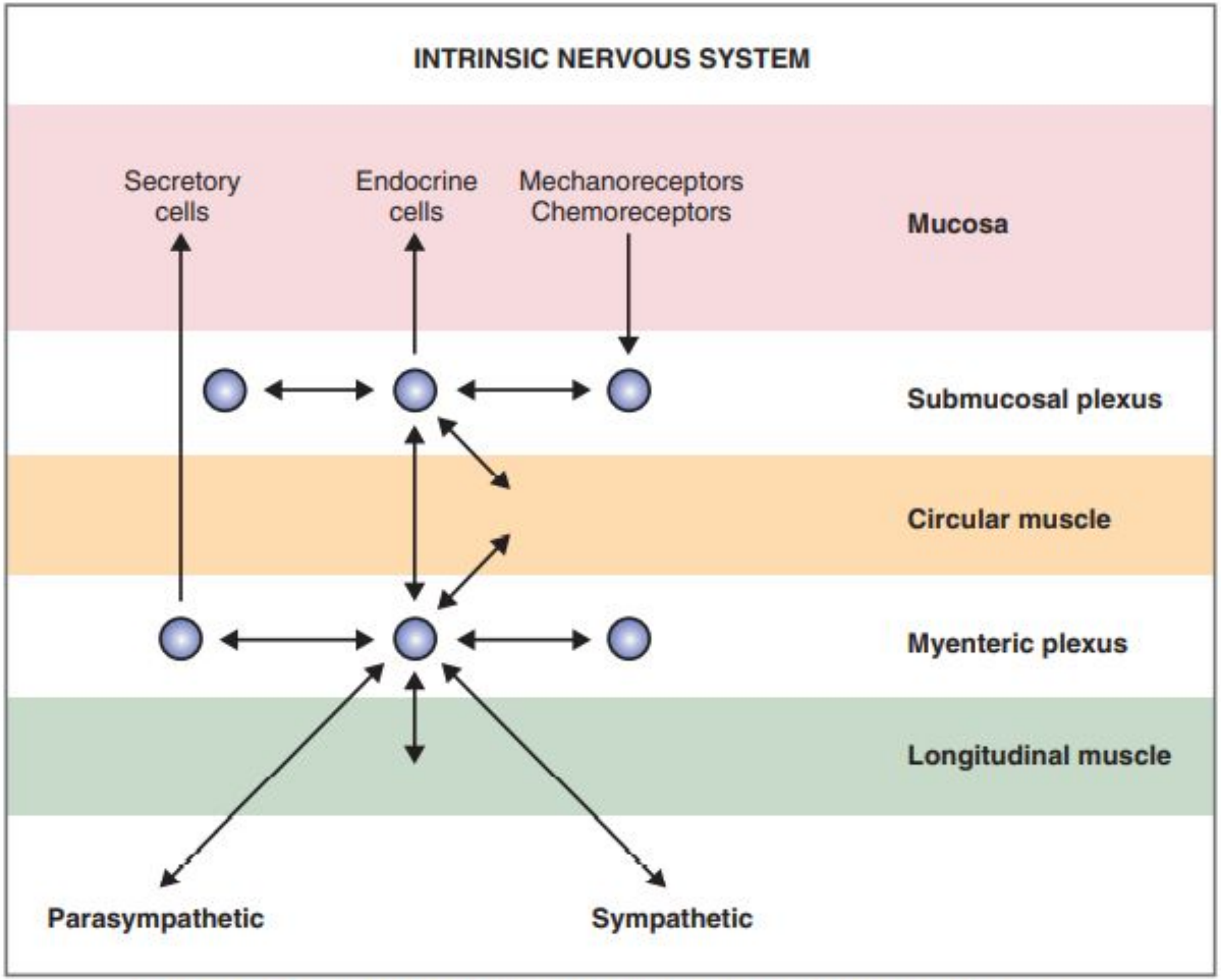

Figure 8–3 Intrinsic nervous system of the gastrointestinal tract.

- These nerve plexus contain nerve cell bodies, processes of nerve cells and the receptors. The receptors in the GI tract are stretch receptors and chemoreceptors. Enteric nervous system is controlled by extrinsic nerve.
- These ganglia receive input from the parasympathetic and sympathetic nervous systems, which modulate their activity. These ganglia also receive sensory information directly from mechanoreceptors and chemoreceptors in the mucosa and send motor information directly to smooth muscle, secretory, and endocrine cells. Information is also relayed between ganglia by interneurons.

Myenteric plexus or Auerbach's plexus

- **Functions of Auerbach plexus** Major function of this plexus is to regulate the movements of GI tract. Some nerve fibers of this plexus accelerate the movements by secreting the excitatory neurotransmitter substances like acetylcholine, serotonin and substance P. Other fibers of this plexus inhibit the GI motility by secreting the inhibitory neurotransmitters such as vasoactive intestinal polypeptide (VIP), neurotensin and enkephalin.

Submucosal plexus or Meissner's plexus

- Meissner plexus is otherwise called submucus nerve plexus. It is situated in between the muscular layer and submucosal layer of GI tract.
- **Functions of Meissner plexus** Function of Meissner plexus is the regulation of secretory functions of GI tract. These nerve fibers cause constriction of blood vessels of GI tract.

EXTRINSIC NERVE SUPPLY

- Extrinsic nerves that control the enteric nervous system are from autonomic nervous system. Both sympathetic and parasympathetic divisions of autonomic nervous system innervate the GI tract.

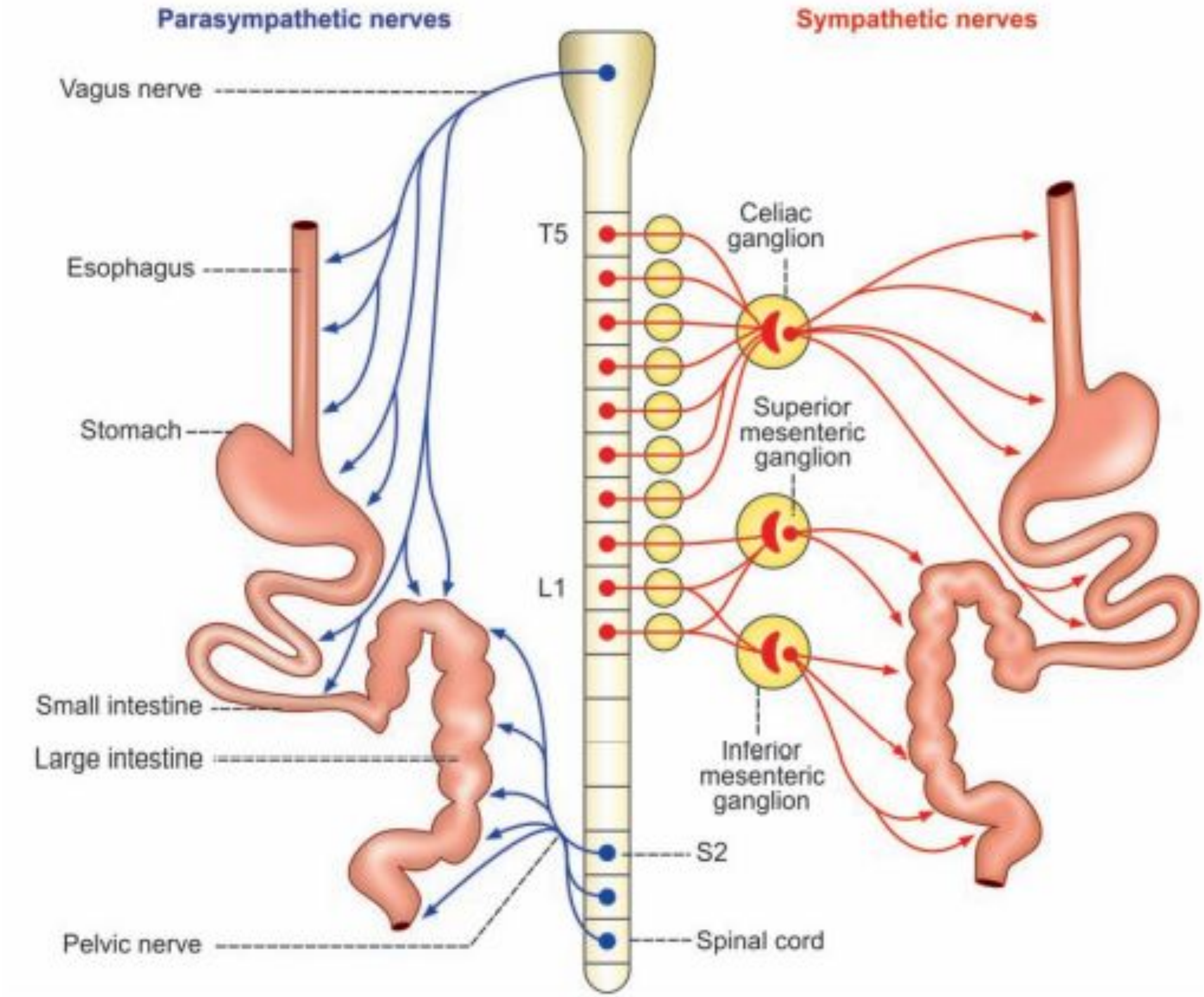

FIGURE 36.3: Extrinsic nerve supply to GI tract.
T5 = 5th thoracic segment of spinal cord
L1 = 1st lumbar segment of spinal cord
S2 = 2nd sacral segment of spinal cord

Sympathetic Innervation

- Preganglionic fibers of the sympathetic nervous system are relatively short and synapse in ganglia outside the gastrointestinal tract.
- Four sympathetic ganglia serve the gastrointestinal tract: celiac, superior mesenteric, inferior mesenteric, and hypogastric .
- Postganglionic nerve fibers, which are adrenergic (i.e., release norepinephrine), leave these sympathetic ganglia and synapse on ganglia in the myenteric and submucosal plexuses, or they directly innervate smooth muscle, endocrine, or secretory cells.

Functions of sympathetic nerve fibers

- Sympathetic nerve fibers inhibit the movements and decrease the secretions of GI tract by secreting the neurotransmitter noradrenaline. It also causes constriction of sphincters.

Parasympathetic Innervation

- Parasympathetic nerve fibers to GI tract pass through some of the cranial nerves and sacral nerves.
- The preganglionic and postganglionic parasympathetic nerve fibers to mouth and salivary glands pass through facial and glossopharyngeal nerves.
- Preganglionic parasympathetic nerve fibers to esophagus, stomach, small intestine and upper part of large intestine pass through vagus nerve. Preganglionic nerve fibers to lower part of large intestine arise from second, third and fourth sacral segments (S2, S3 and S4) of spinal cord and pass through pelvic nerve. All these preganglionic parasympathetic nerve fibers synapse with the postganglionic nerve cells in the myenteric and submucus plexus.

Functions of parasympathetic nerve fibers

- Parasympathetic nerve fibers accelerate the movements and increase the secretions of GI tract. The neurotransmitter secreted by the parasympathetic nerve fibers is acetylcholine (Ach).

Parasympathetic Innervation

- Parasympathetic innervation is supplied by the vagus nerve and the pelvic nerve .
- The vagus nerve innervates the upper gastrointestinal tract including the striated muscle of the upper third of the esophagus, the wall of the stomach, the small intestine, and the ascending colon.
- The pelvic nerve innervates the lower gastrointestinal tract including the striated muscle of the external anal canal and the walls of the transverse, descending, and sigmoid colons.

Parasympathetic Stimulation Increases Activity of the Enteric Nervous System.

- the parasympathetic nervous system has long preganglionic fibers that synapse in ganglia in or near the target organs. In the gastrointestinal tract, these ganglia actually are located in the walls of the organs within the myenteric and submucosal plexuses. Information relayed from the parasympathetic nervous system is coordinated in these plexuses and then relayed to smooth muscle, endocrine, and secretory cells.

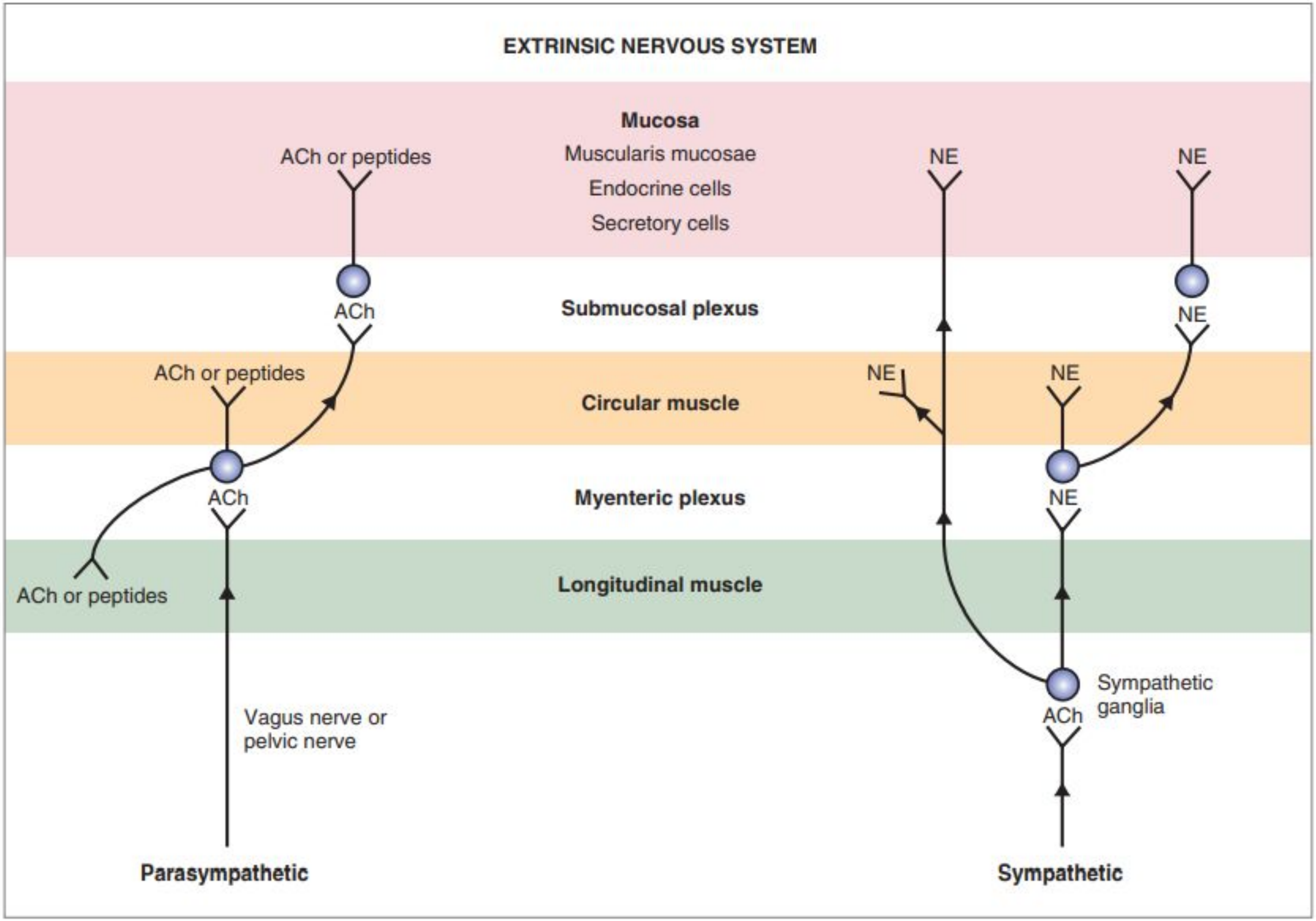

Figure 8–2 The extrinsic nervous system of the gastrointestinal tract. Efferent neurons of the parasympathetic and sympathetic nervous systems synapse in the myenteric and submucosal plexuses, in the smooth muscle, and in the mucosa. ACh, Acetylcholine; NE, norepinephrine.

- Postganglionic neurons of the parasympathetic nervous system are classified as either cholinergic or peptidergic.
- Cholinergic neurons release acetylcholine (ACh) as the neurotransmitter.
- Peptidergic neurons release one of several peptides including substance P and vasoactive inhibitory peptide (VIP).
- The vagus nerve is a mixed nerve in which 75% of the fibers are afferent and 25% are efferent. Afferent fibers deliver sensory information from the periphery (e.g., from mechanoreceptors and chemoreceptors in the wall of the gastrointestinal tract) to the central nervous system (CNS). Efferent fibers deliver motor information from the CNS to target tissues in the periphery (e.g., smooth muscle, secretory, and endocrine cells).
- Thus, mechanoreceptors and chemoreceptors in the gastrointestinal mucosa relay afferent information to the CNS via the vagus nerve, which triggers reflexes whose efferent limb is also in the vagus nerve. Such reflexes, in which both afferent and efferent limbs are contained in the vagus nerve, are called **vagovagal reflexes**

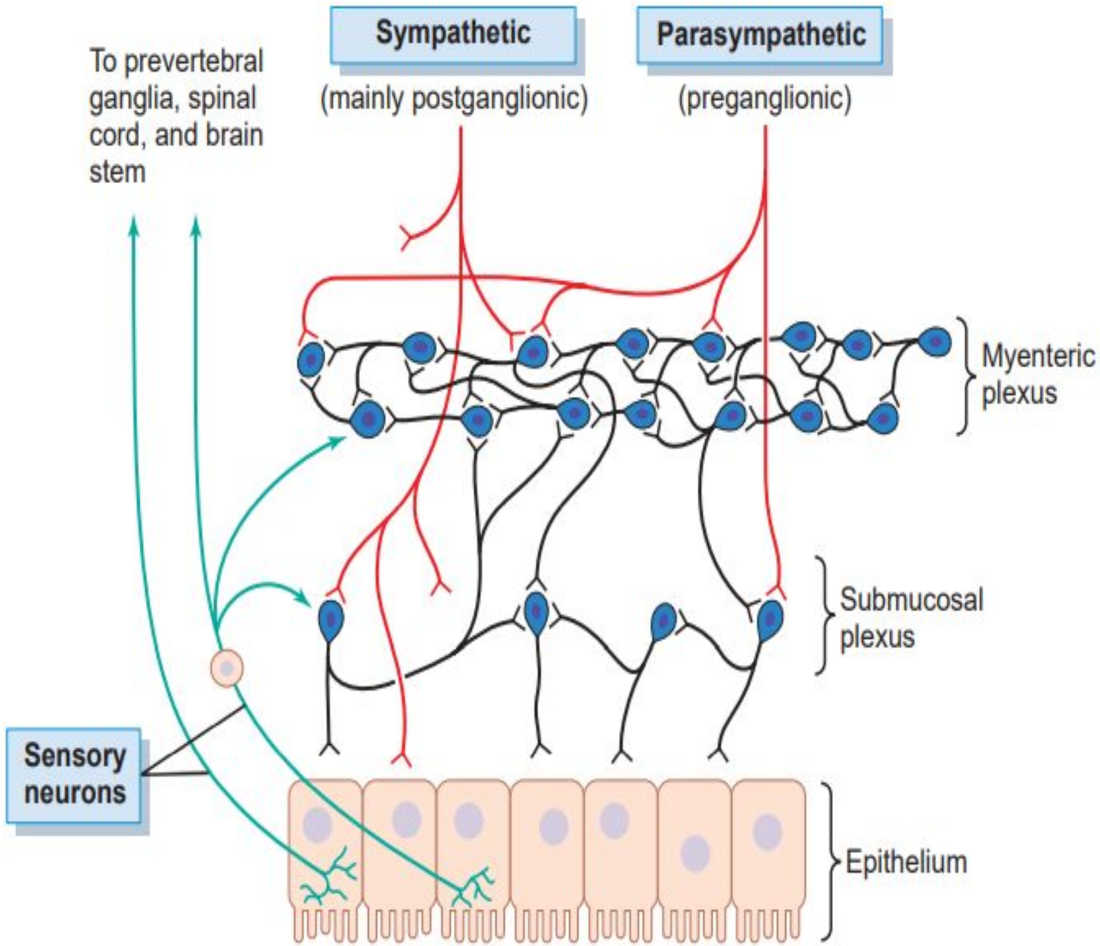

Figure 63-4. Neural control of the gut wall, showing (1) the myenteric and submucosal plexuses (*black fibers*); (2) extrinsic control of these plexuses by the sympathetic and parasympathetic nervous systems (*red fibers*); and (3) sensory fibers passing from the luminal epithelium and gut wall to the enteric plexuses, then to the prevertebral ganglia of the spinal cord and directly to the spinal cord and brain stem (*green fibers*).

DIFFERENCES BETWEEN THE MYENTERIC AND SUBMUCOSAL PLEXUSES

- The myenteric plexus consists mostly of a linear chain of many interconnecting neurons that extends the entire length of the gastrointestinal tract.
- As the myenteric plexus extends all the way along the intestinal wall and lies between the longitudinal and circular layers of intestinal smooth muscle, it is concerned mainly with controlling muscle activity along the length of the gut. When this plexus is stimulated, its principal effects are (1) increased tonic contraction, or “tone,” of the gut wall; (2) increased intensity of the rhythmical contractions; (3) slightly increased rate of the rhythm of contraction; and (4) increased velocity of conduction of excitatory waves along the gut wall, causing more rapid movement of the gut peristaltic waves.

- The myenteric plexus should not be considered entirely excitatory because some of its neurons are inhibitory;
- their fiber endings secrete an inhibitory transmitter, possibly vasoactive intestinal polypeptide or some other inhibitory peptide. The resulting inhibitory signals are especially useful for inhibiting some of the intestinal sphincter muscles that impede movement of food along successive segments of the gastrointestinal tract, such as the pyloric sphincter, which controls emptying of the stomach into the duodenum, and the sphincter of the ileocecal valve, which controls emptying from the small intestine into the cecum.

- The submucosal plexus, in contrast to the myenteric plexus, is mainly concerned with controlling function within the inner wall of each minute segment of the intestine. For instance, many sensory signals originate from the gastrointestinal epithelium and are then integrated in the submucosal plexus to help control local intestinal secretion, local absorption, and local contraction of the submucosal muscle that causes various degrees of infolding of the gastrointestinal mucosa.

- The intrinsic or enteric nervous system can direct all functions of the gastrointestinal tract, even in the absence of extrinsic innervation. The enteric nervous system is located in ganglia in the myenteric and submucosal plexuses and controls the contractile, secretory, and endocrine functions of the gastrointestinal tract. As these ganglia receive input from the parasympathetic and sympathetic nervous systems, which modulate their activity. These ganglia also receive sensory information directly from mechanoreceptors and chemoreceptors in the mucosa and send motor information directly to smooth muscle, secretory, and endocrine cells. Information is also relayed between ganglia by interneurons

- A large number of neurochemicals, or neurocrines, have been identified in neurons of the enteric nervous system . Some of the substances are classified as neurotransmitters and some are neuromodulators (i.e., they modulate the activity of neurotransmitters). Most neurons of the enteric nervous system contain more than one neurochemical, and upon stimulation, they may cosecrete two or more neurocrines.


THANK YOU!